

What tests and measures accurately diagnose persisting post-concussive symptoms in children, adolescents and adults following sport-related concussion? A systematic review

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ABSTRACT

Objective To determine what tests and measures accurately diagnose persisting post-concussive symptoms (PPCS) in children, adolescents and adults following sport-related concussion (SRC).

Design A systematic literature review.

Data sources MEDLINE, Embase, PsycINFO, Cochrane Central Register of Controlled Trials, CINAHL and SPORTDiscus through March 2022.

Eligibility criteria Original, empirical, peer-reviewed findings (cohort studies, case-control studies, cross-sectional studies and case series) published in English and focused on SRC. Studies needed to compare individuals with PPCS to a comparison group or their own baseline prior to concussion, on tests or measures potentially affected by concussion or associated with PPCS. Results: Of 3298 records screened, 26 articles were included in the qualitative synthesis, including 1016 participants with concussion and 531 in comparison groups; 7 studies involved adults, 8 involved children and adolescents and 11 spanned both age groups. No studies focused on diagnostic accuracy. Studies were heterogeneous in participant characteristics, definitions of concussion and PPCS, timing of assessment and the tests and measures examined. Some studies found differences between individuals with PPCS and comparison groups or their own pre-injury assessments, but definitive conclusions were not possible because most studies had small convenience samples, cross-sectional designs and were rated high risk of bias.

Conclusion The diagnosis of PPCS continues to rely on symptom report, preferably using standardised symptom rating scales. The existing research does not indicate that any other specific tool or measure has satisfactory accuracy for clinical diagnosis. Future research drawing on prospective, longitudinal cohort studies could help inform clinical practice.

INTRODUCTION

Up to 30% of children and adults who sustain concussions will experience persisting post-concussion symptoms (PPCS).^{1–3} The fifth International Conference on Concussion in Sport defined

WHAT IS ALREADY KNOWN?

- ⇒ Up to 30% of children and adults with sport-related concussion experience persisting post-concussive symptoms (PPCS).
- ⇒ A variety of pre-injury, acute and subacute factors, both neurobiological and psychosocial in nature, can help predict PPCS.
- ⇒ Tests and measures validated for the contemporaneous diagnosis of PPCS are needed.

WHAT ARE THE NEW FINDINGS?

- ⇒ Of 26 studies that met inclusion criteria, none focused on diagnostic accuracy. Most had small convenience samples, cross-sectional designs and a high risk of bias. Studies also were heterogeneous regarding participant characteristics, the definitions of concussion and PPCS, timing of assessment and the tests and measures examined.
- ⇒ The diagnosis of PPCS continues to depend on symptom report, preferably using standardised and validated symptom rating scales. The scant existing research does not indicate that any other specific tool or measure has satisfactory accuracy for the clinical diagnosis of PPCS.
- ⇒ Large prospective, longitudinal cohort studies are needed to examine the diagnosis of PPCS using a common set of tests and measures. Future studies would benefit from shared operational criteria for the diagnosis of both concussion and PPCS.

persisting post-concussive symptoms (PPCS) as symptoms lasting 2 weeks or longer for adults and 4 weeks or longer for children after a sport-related concussion (SRC).^{4,5} PPCS are frequently associated with mental health problems, declines in quality of life and difficulties returning to sport, school, work and other everyday activities.^{6,7}

A variety of pre-injury, acute and subacute factors, both neurobiological and psychosocial in nature, have been identified that can contribute to



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the prediction of PPCS after SRC.⁸ However, factors that are prognostic of PPCS may not be useful weeks to months later for diagnosing PPCS, documenting impairments associated with PPCS or guiding treatment. Identifying the biopsychosocial features that characterise individuals with PPCS will help deepen our scientific understanding of persisting symptoms and assist clinically with differential diagnosis and treatment planning.⁹

Most tests and measures used to assess individuals presenting with PPCS were originally validated for the assessment and diagnosis of concussion in the acute or subacute period.¹⁰ However, evidence for their use in the contemporaneous assessment and diagnosis of PPCS is limited. A previous systematic review conducted as part of 5th International Conference on Concussion in Sport identified only 11 articles focused on the assessment of PPCS, with low overall quality and moderate risk of bias (ROB) (ie, all studies were cross-sectional or case series).⁵ Despite the paucity of direct evidence, the authors concluded that PPCS does not reflect a single pathophysiological entity or necessarily reflect ongoing physiological injury to the brain, but instead involves non-specific symptoms that may be linked to concussion but also to pre-existing or co-occurring neurodevelopmental conditions, health problems or psychosocial factors. A detailed multimodal clinical assessment was recommended to identify specific primary and secondary processes that may contribute to persisting symptoms.

Building on these findings, the current systematic review sought to determine what tests and measures accurately diagnose PPCS in children, adolescents and adults following SRC. The review focused on original, empirical, peer-reviewed research on SRC that compared individuals with PPCS to a comparison group or their own baseline prior to concussion. Based on the available evidence, the review also considered two pertinent clinical questions: should the assessment of PPCS be different than that recommended for assessment of SRC in the acute phase? What differential diagnoses should be considered in the presence of PPCS?

METHODS

The review protocol was prospectively registered in the PROSPERO database for systematic reviews (protocol ID: CRD42020153730) and is reported here according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) and PERSiST guidelines.^{11 12} The review included articles that compared individuals with PPCS after SRC to a comparison group (ie, individuals without PPCS after concussion, healthy individuals without concussion or individuals with injuries other than concussion) or to their own baseline prior to concussion on tests or measures potentially affected by concussion or linked to PPCS. At the editor's request, we amended our original study protocol to rate the overall quality of the body of evidence and any associated recommendations using the Strength of Recommendation Taxonomy (SORT), which entails three levels: consistent and good-quality patient-oriented evidence (level 1); inconsistent or limited-quality patient-oriented evidence (level 2) and consensus, usual practice and/or opinion (level 3).¹³

Search strategy

The search strategy focused on three main concepts: concussion, sports and persisting symptoms. Content experts on the research team provided keywords and seed articles from which the search strategies were developed. A standard search focusing on the concepts of concussion and sports was developed in MEDLINE

by two expert health science librarians.¹⁴ The search strategy was composed of subject headings and keyword terms. This search was peer-reviewed with PRESS¹⁵ by another experienced health sciences librarian. The third concept focused on persisting symptoms. The search strategy for this concept was peer reviewed by Dr KA Hayden, Librarian, University of Calgary. The penultimate search strategy was pilot tested with seed articles, as well as relevant studies included in systematic reviews from the 5th International Conference on Concussion in Sport. The search strategy was shared with the lead author for feedback, and adjustments were made if required. The final search strategy was translated for all other databases. The subject headings were responsive to the controlled vocabulary of each database, and keywords remained the same for each database. The following databases were searched: MEDLINE (OVID); Embase (OVID); PsycINFO (OVID); Cochrane Central Register of Controlled Trials (OVID); CINAHL Plus with Full Text (EBSCO) and SPORTDiscus with Full Text (EBSCO). The full search strategy is presented in online supplemental material. The searches were initially completed for the period 2001–12 October 2019, and then updated by re-running the search strategies in each database through 22 March 2022. Search results were uploaded to, and deduplicated in, Covidence (Veritas Health Innovation, Melbourne, Australia; <https://www.covidence.org/>).

Article screening and selection

One author (AMR) performed a rapid screen to exclude citations that were clearly not related to SRC, did not involve human participants or were not original research. Each remaining title/abstract was screened by two authors independently. Discrepancies were resolved by the lead author. If no consensus was reached, the citation was included and further reviewed in full text. Each full text was reviewed by two authors independently for possible inclusion. Discrepancies were resolved by the lead author. To establish inter-rater agreement among all authors involved in title/abstract and full text screening, a random sample of 50 Medline records was exported to an Excel spreadsheet by the librarian not involved in the screening (ZP). Authors independently screened the 50 records and returned their file to the librarian. Inter-rater agreement was calculated by comparing the ratings of the authors who screened a subsample of the records to those of the author who screened all records (AMR). If inter-rater agreement was <80%, another 50 records were screened after discussing discrepancies. Inter-rater agreement was established again for the updated search in June 2022.

To be included in the review, articles needed to meet the following inclusion criteria: (1) original research (cohort studies, case-control studies, cross-sectional studies and case series); (2) SRC as the primary source of injury (ie, at least 50% of mechanism must be sport-related or results for sport-related subgroup reported separately); (3) human participants; (4) assess PPCS (ie, post-concussive symptoms lasting >2 weeks in adults aged 19 years and older and >4 weeks in children aged 5–12 years and adolescents aged 13–18 years; all participants must have been assessed beyond these cut-offs, and it was not sufficient if only the group mean or median time post injury was longer); (5) report the outcome of specific tests or measures potentially associated with concussion or PPCS; and (6) English language. Studies were excluded for meeting any of the following criteria: (1) review articles, case studies and intervention studies (except studies reporting relevant baseline pre-treatment comparisons); (2) studies published only in abstract form; (3) focus on non-sport-related mechanism of injury (eg, falls, assault, motor vehicle

collisions and blast injuries); (4) tests or measures completed at <2 weeks for adults and <4 weeks for children and adolescents following concussion; (5) include only asymptomatic individuals (ie, those who are no longer symptomatic or are considered symptomatically recovered following concussion), or do not distinguish those with PPCS from those without PPCS.

Data extraction

Two authors extracted data for each study using predesigned forms (available from the first author), and AMR resolved any disagreements. The following data were extracted from each included manuscript: (1) study design; (2) authors; (3) year and location of study; (4) study duration; (5) participant demographics, including number of participants (n) by age, sex, type of sport and level of sport; (6) definition of concussion; (7) definition of PPCS; (8) test or measures included in the study; (9) comparison group(s) and (10) results (including statistical outcomes, with point estimate and measure of precision when available or able to be computed based on data presented). Because of the small number of studies and heterogeneity in study methods, especially outcome measures, no meta-analysis was performed, and the results were synthesised in a narrative review.

ROB assessment

ROB assessment was performed by two authors using the appropriate Scottish Intercollegiate Guidelines Network (SIGN; <https://www.sign.ac.uk/>) critical appraisal notes and checklist. Although the SIGN checklist for diagnostic accuracy studies would have been most appropriate for this review, no included studies were explicitly framed as diagnostic accuracy studies, so the checklists for cohort or case-control studies were used instead. All articles were rated unacceptable (high ROB), acceptable (moderate ROB) or high quality (low ROB) based on the assessment. Disagreements between reviewers were resolved by a third author (KOY or AMR). ROB rating templates are available in the supplemental material of the consensus methodology paper.¹⁴

Equity, diversity and inclusion statement

We included all eligible studies in the systematic review regardless of sample characteristics, including sex, gender, race/ethnicity, socioeconomic level or representation from marginalised groups. Our data extraction forms did not include any of these characteristics other than sex, but we subsequently reviewed all included studies to determine if they reported on gender, race, ethnicity or other social determinants of health and comment on this in the Discussion section. The authors of this review include 7 women and 10 men from a variety of disciplines and a range of career stages, including a student and two early career researchers, who were predominantly from North America but also from Australia and Europe. Based on self-disclosures, the team includes persons of colour but lacks the perspectives of members of the 2SLGBTQI+ community or persons from low-income and middle-income geographical regions.

RESULTS

From 3298 records screened, 26 articles were included in the qualitative synthesis.^{16–41} Figure 1 presents a modified PRISMA diagram of the study screening and selection process. Table 1 provides a summary of study characteristics, and the data extracted about each individual study and its ROB rating are presented in the online supplemental material.

The 26 articles included 366 separate outcome measures, which often included multiple metrics derived from the same measurement tools or instruments (eg, multiple metrics from numerous regions of interest in a study of diffusion tensor imaging). Many papers examined multiple tools or instruments. We classified the tools and instruments into 17 major categories, as shown in table 2, divided between those involving neuroimaging versus other types of tests and measures. Most studies (58%) were rated unacceptable/high ROB, with the remainder rated acceptable/moderate ROB; none were rated high quality/low ROB. Figure 2 shows the distribution of ROB ratings in each of the 17 outcome categories, based on only moderate (acceptable) and high (unacceptable) ROB ratings.

The studies included a total of 1016 participants with concussion, with a median sample size of 28, and a total of 531 participants in comparison groups, with a median sample size of 17. Across the 26 articles, 7 involved adults only,^{20 23 33 36–39} 8 involved children and adolescents only,^{16 22 27–30 35 40} and 11 blended both age groups, most often including both adolescents and adults.^{17–19 21 24–26 31 32 34 41} (the age range for studies that did not report their participants' full age range was approximated as mean±2SD). Participants in both concussion and comparison groups were largely male; four studies included only males^{18–20 36} and one included only females.²¹ Most studies (73%) were conducted in North America.

No studies focused on diagnostic accuracy. All studies but one²⁸ involved cross-sectional designs, mostly comparing individuals with PPCS to healthy individuals, although some studies involved comparisons to individuals with concussion who were asymptomatic (ie, no longer symptomatic) at the time of assessment. Some studies included repeated assessments post injury; only outcomes measured >2 weeks after injury in adults and >4 weeks in children and adolescents are reported. Participants were often recruited from specialty concussion clinics. Most studies (77%) included only participants with PPCS who sustained sport-related or recreation-related concussions, arising in a variety of contact and collision sports at both amateur/recreational and elite/professional levels. Some studies did not report details on the specific sports, activities or settings involved.

Only one study provided clear operational criteria for concussion diagnosis.⁴⁰ Some studies referenced the International Consensus on Concussion in Sport,⁴² American Congress of Rehabilitation Medicine⁴³ or American Academy of Neurology⁴⁴ definitions of concussion or mild traumatic brain injury. Some studies relied on physician diagnosis without any criteria, and others did not provide a clear description of how concussion was diagnosed. The identification of PPCS was most often based solely on self-report of persisting symptoms, and sometimes focused on specific symptoms (eg, depression) rather than a broader range of possible symptoms. Some studies referred to the International Classification of Diseases⁴⁵ or Diagnostic and Statistical Manual-4th Edition⁴⁶ definitions of PPCS. Seven studies offered specific definitions of PPCS that included quantitative criteria based on symptom rating scales.^{24–28 32 40} Regardless of the definition used, the specific symptoms that characterised participants with PPCS were not typically reported.

Summary of study results

Study results are summarised here and described in greater detail in the online supplemental material. Neuroimaging outcomes were examined in 15 studies.^{16 18–20 23 24 27 29–34 40 41} They revealed few if any structural abnormalities in individuals with PPCS,^{16 18–20 29–34} as well as few robust volumetric

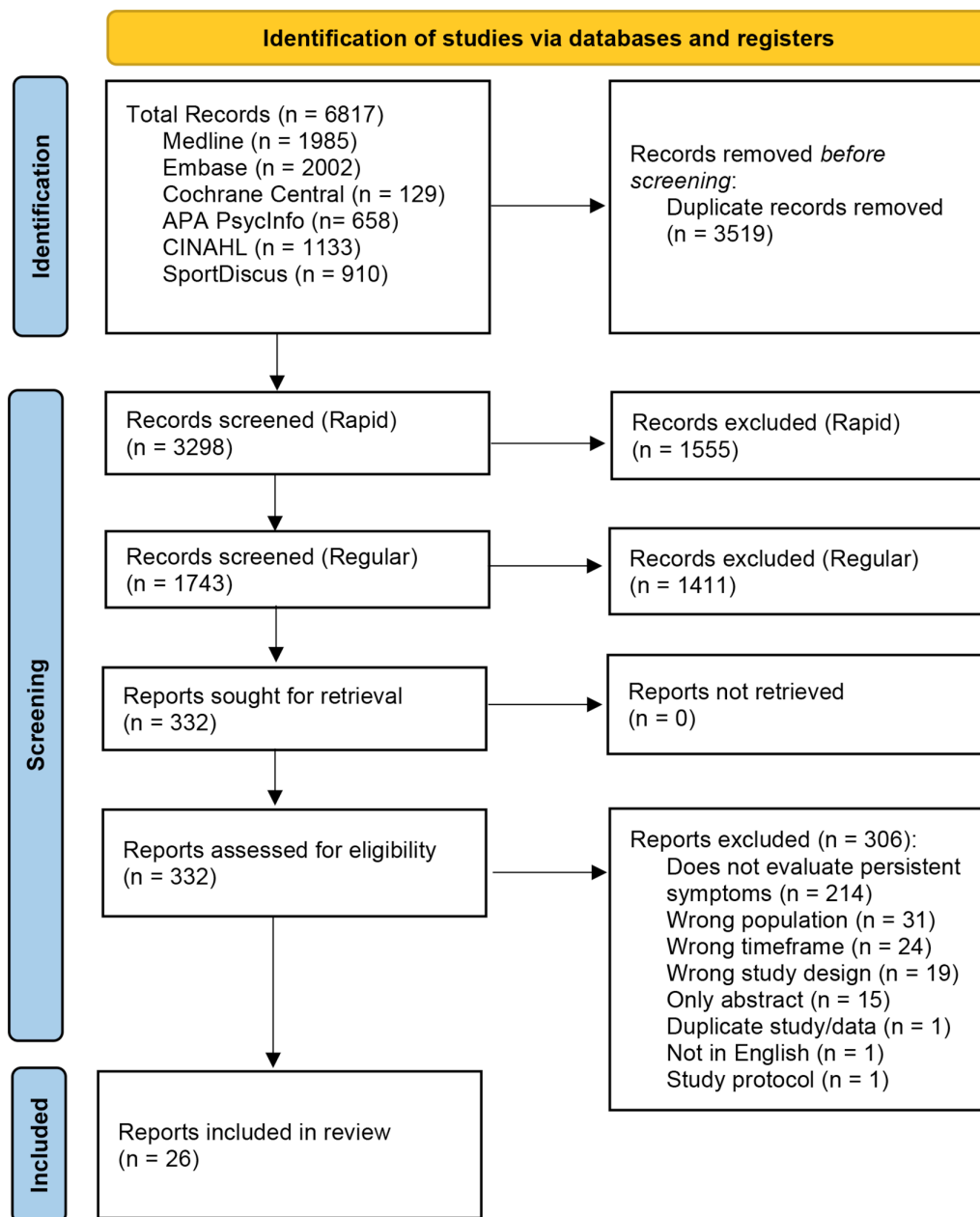


Figure 1 Modified Preferred Reporting Items for Systematic Reviews and Meta-Analyses diagram of the study screening and selection process.

differences.^{19 23 29–31} Studies using diffusion imaging found some evidence of atypical microstructure in select white matter tracts, often in more anterior brain regions.^{16 23 30 34} Findings regarding cerebral blood flow and cerebrovascular reactivity were inconsistent, and some differences associated with PPCS appeared to be patient specific.^{16 32 40} Studies of task-based functional MRI provided evidence for reduced prefrontal activation on working memory tasks and corresponding increased activation (or attenuated deactivation) in other regions.^{18–20 27} No robust findings emerged in a single study of magnetic resonance spectroscopy with a small sample.¹⁶ A single positron emission tomography study found evidence of increased tau aggregation and neuroinflammation in individuals with PPCS, but the sample comprised a very small group of individuals with multiple SRC.³¹ A single study of functional near-infrared spectroscopy provided an additional indication of task-based reductions in frontal brain activity.²⁴ Finally, a single study using multimodal MRI suggested that advanced data fusion techniques and machine learning may

yield improved diagnostic accuracy for PPCS, although accuracy was lower for individuals with only one concussion or at longer times post injury.⁴¹

Other tests and measures, aside from imaging, were examined in 17 studies.^{17 21–31 35–39} The studies provide some indication that PPCS is associated with differences from comparison groups on neurophysiological measures (ie, increased cortical excitability and slower event-related potentials),^{28 39} fluid biomarkers (ie, increased neurofilament light and astroglial activation),^{31 36–38} vestibular and balance measures,^{22–24 35} visual/ocular measures,^{17 22 29 30} respiratory and vascular measures (eg, altered cerebral blood flow and CO₂ sensitivity),²¹ cognitive tests (eg, poorer working memory and slower reaction time)^{27 29–31 39} and rating scale measures (eg, more mood problems and poorer quality of life).^{23 29 30 39} However, no more than four studies were included in any specific outcome category, and the studies within specific categories usually examined different tests or measures.

Table 1 Summary of study characteristics

Participants in concussion groups	Total n across 26 studies 1016 Median n per study 28, IQR 39.5, range 8–127 Median % male per study 57, IQR 33, range 0–100
Participants in healthy comparison groups	Total n across 26 studies 531 Median n per study 17, IQR 13.0, range 6–58 Median % male per study 58, IQR 33, range 0–100
% participants with sport-related concussion per study	Median 100, IQR 6, range 50–100
Age range of participants in study	
Children and adolescents only	8 studies (31%)
Adults only	7 studies (27%)
Both	11 studies (42%)
Location of study	
USA	11 studies (42%)
Canada	8 studies (31%)
Sweden	4 studies (15%)
Germany	3 studies (12%)
Study design	
Cohort	1 study (4%)
Cross-sectional	25 studies (96%)
Concussion definition used in study	
Operational criteria	1 study (4%)
Organisational definition	11 studies (42%)
Physician diagnosis	3 studies (16%)
Self-report	1 study (4%)
Not provided	10 studies (38%)
PPCS definition used in study	
Symptom rating scale criteria	5 studies (19%)
Organisational definition	7 studies (27%)
Self-report	13 studies (50%)
Physician report	1 study (4%)
High risk of bias in study	15 studies (58%)

Figure 3 provides a high-level summary of all comparisons made across the 17 outcome categories in the 26 included studies. The figure shows the total number of metrics included in each category, including all variants or subscales of specific measures, as well as the total percentage of statistical comparisons that were significant relative to the total number of comparisons. As the figure highlights, <40% of the statistical comparisons were significant in all but three outcome categories, and those with higher proportions of significant comparisons were often based on a small number of metrics and comparisons. Given the small number of studies in most outcome domains, the lack of any high-quality studies, and the inconsistent or limited results, the overall body of evidence and associated recommendation regarding the use of specific tests or measures in the clinical diagnosis of PPCS were judged to be level two according to SORT (ie, inconsistent or limited quality patient-oriented evidence).¹³

DISCUSSION

This systematic review sought to determine what tests and measures can support the diagnosis of PPCS in children, adolescents and adults following SRC. The review identified 26 studies that met the specified inclusion/exclusion criteria.^{16–41} Importantly, no studies were explicitly designed to assess diagnostic accuracy. The studies were heterogeneous in participant characteristics, definitions of concussion and PPCS, timing of

Table 2 Major outcome categories and associated studies

Outcome category	Studies
Imaging	
Structural abnormalities	Bartnik-Olson et al 2014 ¹⁶ Chen et al 2007 ¹⁸ Chen et al 2008b ²⁰ MacDonald et al 2019 ³⁰ Marklund et al 2021 ³¹ Mutch et al 2016 ³² Panwar et al 2020 ³³ Polak et al 2015 ³⁴
Quantitative volumetrics	Gard et al 2022 ²³ MacDonald et al 2019 ³⁰ Marklund et al 2021 ³¹
Diffusion imaging	Bartnik-Olson et al 2014 ¹⁶ Gard et al 2022 ²³ MacDonald et al 2019 ³⁰ Polak et al 2015 ³⁴
Cerebral blood flow/Cerebrovascular reactivity	Bartnik-Olson et al 2014 ¹⁶ Mutch et al 2016 ³¹ Stephens et al 2018 ⁴⁰
Task based functional MRI	Chen et al 2007 ¹⁸ Chen et al 2008a ¹⁹ Chen et al 2008b ²⁰ Khetani et al 2019 ²⁷
Magnetic resonance spectroscopy	Bartnik-Olson et al 2014 ¹⁶
Positron emission tomography	Marklund et al 2021 ³¹
Functional near infrared spectroscopy	Helmich et al 2020a ²⁴
Multimodal MRI	Tamez-Pena et al 2022 ⁴¹
Non-imaging	
Neurophysiological measures	King et al 2019 ²⁸ Sicard et al 2021 ³⁹
Fluid biomarkers	Marklund et al 2021 ³¹ Shahim et al 2016 ³⁶ Shahim et al 2017 ³⁷ Shahim et al 2020 ³⁸
Vestibular/balance measures	Cochrane et al 2021 ²² Gard et al 2022 ²³ Helmich et al 2020a ²⁴ Rocheffort et al 2017 ³⁵
Visual/ocular measures	Cheever et al 2018 ¹⁷ Cochrane et al 2021 ²² MacDonald et al 2018 ²⁹ MacDonald et al 2019 ³⁰
Motor control measures	Helmich et al 2020b ²⁵ Helmich et al 2020c ²⁶
Respiratory/vascular measures	Clausen et al 2016 ²¹
Cognitive tests	Khetani et al 2019 ²⁷ MacDonald et al 2018 ²⁹ MacDonald et al 2019 ³⁰ Marklund et al 2021 ³¹ Sicard et al 2021 ³⁹
Rating scales	Gard et al 2022 ²³ MacDonald et al 2018 ²⁹ MacDonald et al 2019 ³⁰ Sicard et al 2021 ³⁹

assessment post injury and tests and measures examined. A relatively small number of studies focused on any particular outcome. These factors precluded any meta-analysis of study results.

Qualitative synthesis of studies of neuroimaging outcomes pointed to subtle differences in white matter microstructure and brain activation on working memory measures, often in anterior brain regions, that may have potential diagnostic significance for PPCS, although the differences may vary from person to person. Studies of outcomes other than neuroimaging also point

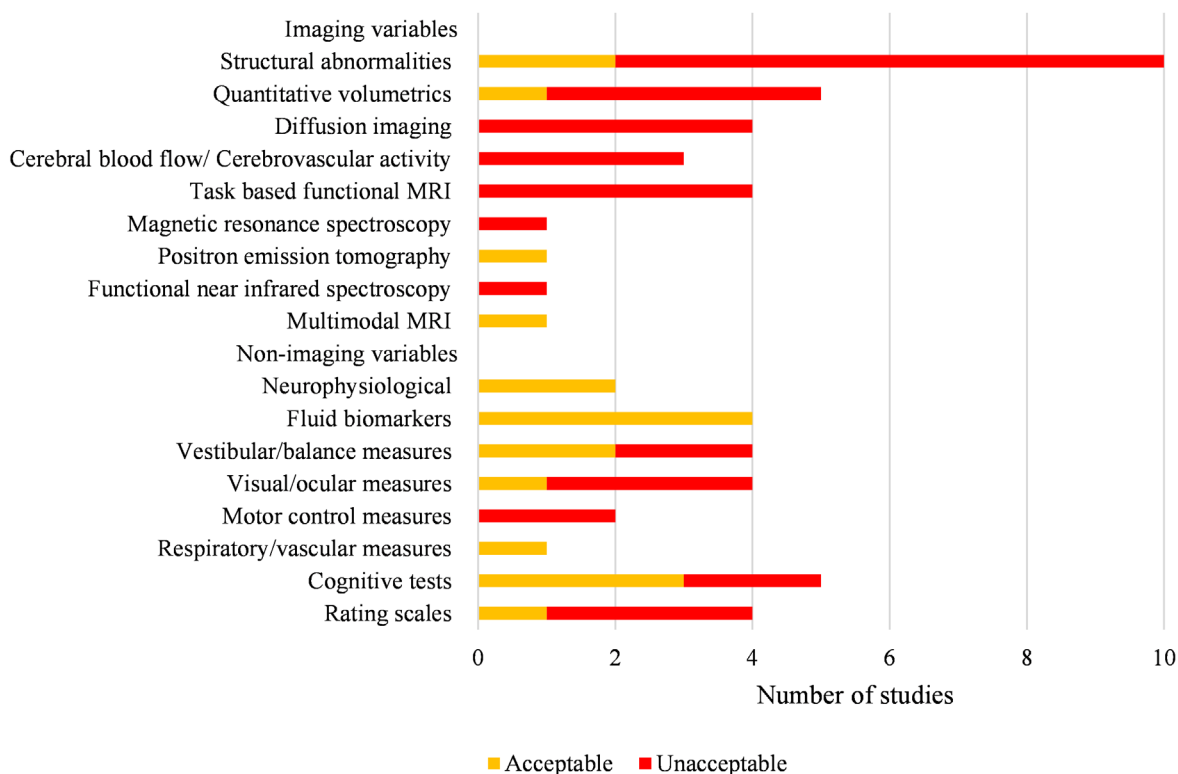


Figure 2 Risk of bias rating by outcome category. Studies could report outcomes in several categories. None of the included studies were assessed to be 'high quality'.

to differences in a variety of domains, ranging from neurophysiological to psychosocial, that may support diagnosis of PPCS. However, the findings in most of the 17 major categories listed

in table 1 were based on a small number of studies, which were themselves heterogeneous in methods and results. Moreover, as figure 3 highlights, most comparisons in each of the categories

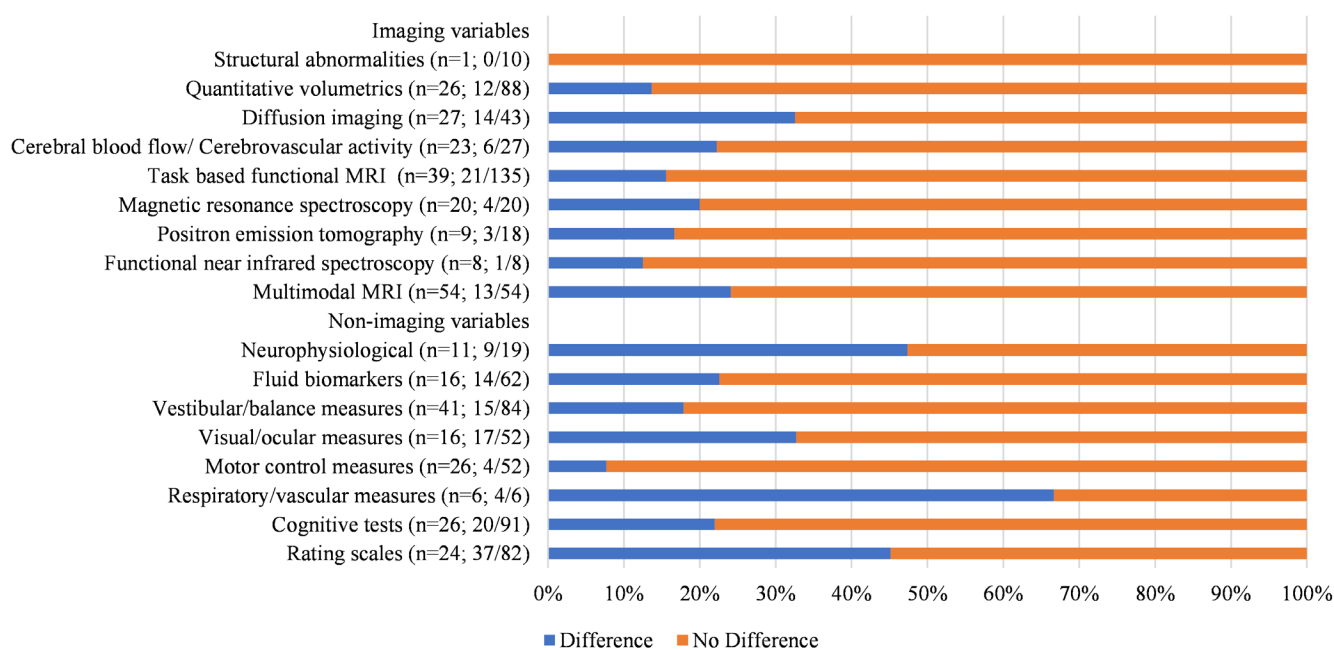


Figure 3 Proportions of statistically significant versus non-significant group comparisons reported in the included studies. 'n' refers to the number of metrics (including subscales) included in each category and is followed by the number of significant group comparisons out of all reported group comparisons.

were not statistically significant, raising concerns about the robustness of the findings.

Despite some differences in a variety of outcome domains between individuals with and without PPCS, evidence-based recommendations regarding the use of specific tools or measures other than standardised symptom rating scales in the clinical diagnosis of PPCS are not currently possible. No studies were rated high quality, and most were rated as having a high ROB, even when using SIGN checklists other than those for diagnostic accuracy studies. The included studies were generally characterised by cross-sectional designs and small convenience samples, often with substantial selection bias (eg, patients recruited from specialised concussion clinics as opposed to inception cohorts that are more likely to be representative of the population of individuals with concussion). Overall, the body of evidence was judged inconsistent and of limited quality (ie, level 2). Because of the shortcomings of the evidence base, a critical unresolved question is how the diagnosis of PPCS might differ for children, adolescents and adults.

As noted already, none of the included papers were clearly intended as diagnostic accuracy studies. Most studies simply compared individuals with PPCS to healthy individuals on specific tests or measures, addressing a phase 1 question according to the diagnostic architecture set forth by Sackett and Haynes (ie, do test results in patients with the target disorder differ from those in typical people?).⁴⁷ Only two studies of acceptable quality^{38 41} attempted to ask whether individuals with specific test results are more likely to experience PPCS than those without such results or whether specific tests can distinguish individuals with vs without PPCS among samples suspected of experiencing PPCS (ie, phases 2 and 3 diagnostic questions per Sackett and Haynes). Those two studies reported on the sensitivity and specificity of certain tests or measures in diagnosing PPCS using receiver operating curve analyses, but neither reported positive predictive values or negative predictive values.

Ideally, diagnostic accuracy studies would involve independent, blind comparisons of test results with a gold standard diagnosis among a consecutive series of patients suspected (but not known) to have PPCS (eg, individuals at high risk per a clinical prediction rule). Unfortunately, no gold standard exists for the diagnosis of PPCS. The diagnosis of PPCS depends on symptom report, preferably assessed using standardised and validated symptom rating scales (eg, graded symptom checklist in the SCAT6;⁴⁸ Health and Behaviour Inventory, contained in the Child SCAT6),⁴⁹ but a shared operational definition of PPCS is lacking in the field. The definition of PPCS varies widely across studies, and different criteria for PPCS have been shown to give rise to widely different estimates of prevalence.^{50 51} To move the field forward, and ensure consistency in research and clinical management, an accepted operational definition of PPCS is needed, as is recognition that PPCS may be expressed as a variety of phenotypes that vary in severity and distribution of symptoms.^{52 53}

Another shortcoming of the existing research is that most studies occurred in North America and comprised predominantly male samples. Most studies did not report on gender, race, ethnicity or other social determinants of health in their samples, all of which can affect participant recruitment and retention.⁵⁴ This puts severe constraints on the generalisability of the findings from the limited research to date. Future research needs to consider sampling adequacy in study design, provide more demographic information about study participants and examine whether the association of PPCS with certain tests or measures is moderated by demographic or socioeconomic factors.

Optimally, future research also would entail large prospective cohort studies of representative samples, with longitudinal follow-up, limited attrition and common data elements when selecting measures of symptoms and outcomes. Researchers should use common operational criteria for concussion. The International Conference on Concussion in Sport definition of concussion was referenced in many included studies but lacks clear operational criteria.⁴² A recent update of the American Congress of Rehabilitation Medicine definition of mild traumatic brain injury provides specific operational criteria that could be used in SRC research.^{55 56} Researchers also should use an accepted operational definition of PPCS, preferably one based on validated rating scales and standardised change in symptom scores (eg, reliable change).⁵¹ However, research is needed to determine whether to use an overarching definition of PPCS that reflects total symptom burden or to define PPCS within specific symptom domains or in terms of symptom trajectories or subtypes, given evidence for distinct symptom clusters and clinical phenotypes that may have different aetiologies and time courses and are likely to differ on specific tests and measures.^{52 53 57 58} Comparison groups likely should not be healthy individuals, but individuals with concussion who do not demonstrate PPCS (ie, those who are no longer symptomatic or whose symptoms do not meet study criteria for PPCS) or individuals who have sustained injuries not involving the head. In the context of SRC, comparison groups should control for important covariates such as sport played, playing position, level of play and other sport-related factors that may influence symptoms. Several large, high-quality studies across the desired age ranges already exist that satisfy many of these criteria, but their data have not yet been used to examine what tests or measures differentiate individuals with and without PPCS after concussion.^{59–63} These studies do not all focus exclusively on SRC, but pertinent analyses could be limited to only those participants injured in sport settings.

Clinically, the limited evidence reviewed here does not provide a basis for stating whether the assessment of PPCS should differ from that recommended for assessing concussion in the acute or subacute phase. If the evidence showed that certain tests or measures had utility in diagnosing PPCS, we could compare those tests and measures to those validated for acute or subacute assessment and offer guidance about how assessment might differ across time post injury. Unfortunately, the evidence regarding assessment of PPCS is simply too limited to allow such a comparison. Therefore, consistent with the most recent international consensus statement and associated systematic review^{5 42} (ie, level 3), we recommend a multimodal clinical assessment, ideally by a multidisciplinary team, to characterise individuals with persisting symptoms, including the types, pattern and severity of symptoms and any associated conditions or other factors that may be causing or contributing to them. In the future, though, existing and new tools and measures should be validated specifically for use in diagnosing PPCS. Once those tools and measures are identified, rather than conducting a multimodal assessment in all patients with persisting symptoms, assessment could potentially begin with highly sensitive screening tools that are likely to identify PPCS and then focus more on targeting assessments and treatments to specific symptom clusters and clinical phenotypes. Further research may identify objective biomarkers that help to distinguish these phenotypes.⁶⁴

When assessing individuals with PPCS, clinicians need to be mindful that the symptoms often thought to be associated with concussion are non-specific, being commonly reported by healthy individuals and those with conditions other than concussion, and can be exacerbated by a variety of biopsychosocial factors aside from

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concussion.^{65–67} Clinicians also need to consider other disorders that may have existed prior to injury (although possibly exacerbated by concussion), co-occur with PPCS or mimic PPCS but not be attributable to the concussion per se.^{68 69} The research summarised in the current review does not speak directly to differential diagnosis. We expected to find studies that compared individuals with concussion to those with other disorders, but none were identified. A more specific search on differential diagnosis might have identified relevant studies. In the absence of firm evidence, differential diagnosis should proceed according to relevant disciplinary or professional standards or guidelines in specific clinical or sporting contexts⁷⁰ (ie, level 3). Common considerations in the context of persisting symptoms include mental health issues⁷¹; learning or attention difficulties⁷²; visual, ocular-motor, cervical and vestibular problems^{73–75}; headache and migraine⁷⁶; sleep disturbance⁷⁷; dysautonomia, including orthostatic intolerance and postural orthostatic tachycardia syndrome^{78 79} and pain.⁸⁰ All these problems or disorders can be either pre-existing or new onset. Validated tests and measures are needed that focus on key symptom clusters and patient phenotypes, to assist in differential diagnosis and treatment planning.

Key recommendations

- ⇒ Persisting symptoms can be assessed using standardised and validated symptom rating scales. However, evidence-based recommendations regarding the use of other specific tests or measures in the clinical diagnosis of persisting symptoms in any age group are not possible based on existing research (level 2).
- ⇒ Currently, a multimodal clinical assessment, ideally by a multidisciplinary team, is recommended to characterise individuals with persisting symptoms, including the types, pattern and severity of symptoms, and any associated conditions or other factors that may be contributing (level 3).
- ⇒ Differential diagnosis should proceed according to relevant disciplinary or professional standards in the specific clinical or sporting context (level 3). Important diagnostic considerations in the context of persisting symptoms include distinguishing pre-existing or new-onset mental health issues; learning or attention difficulties; visual, ocular-motor, cervical and vestibular problems; headache and migraine; sleep disturbance; dysautonomia, including orthostatic intolerance and postural orthostatic tachycardia syndrome; and pain.

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Competing interests KOY is editor-in-chief of the journal *Neuropsychology* for which he receives an editorial stipend from the American Psychological Association. He is an unpaid consulting editor for *Archives of Clinical Neuropsychology* and *Journal of Head Trauma Rehabilitation*. He is an unpaid member of the Scientific Advisory Committee for Brain Injury Canada. He is the chair of the Canadian Concussion Network, which is funded by a grant from Canadian Institutes of Health Research (CIHR) to the University of Calgary; he is principal applicant on the grant but receives no income from it. He is a principal investigator (PI) on another grant from CIHR from which he derives no income. He is a co-investigator on research grants from CIHR, the US National Institutes of Health (NIH), Brain Canada Foundation and National Football League Scientific Advisory Board; he derives income only from the grant from NIH. He serves as a member of a CIHR grant review panel for which he receives an honorarium. He receives book royalties from Guilford Press and Cambridge University Press. He has received travel support and honorarium for presentations to multiple organisations. He has served or serves on the following committees/boards for which he receives honorarium: Independent Data Monitoring Committee (IDMC), Care for Post-Concussive Symptoms Effectiveness (CARE4PCS-2) Trial, National Institute for Child Health and Human Development; Observational Study Monitoring Board (OSMB), Approaches and Decisions in Acute Pediatric TBI (ADAPT) Trial, National Institute of Neurological Disorders and Stroke; National Research Advisory Council, National Pediatric Rehabilitation Resource Center, Center for Pediatric Rehabilitation: Growing Research, Education, and Sharing Science (C-PROGRESS), Virginia Tech University. AMR received a stipend for the clerical and administrative aspects of the systematic review. She has received research grants from the Tonal Strength Institute and Alberta Bone and Joint Strategic Clinical Network. ZP has no disclosures. CD was a pharmaceutical funded invited expert panelist (Teva) for CGRP monoclonal antibody treatment in chronic migraine and received a pharmaceutical funded grant for growth hormone deficiency following concussion. She has grants as PI or co-investigator from the Canadian Department of National Defense, CIHR, Tri-council funding New Frontiers, Alberta Provincial government and Foundations for Physical Medicine and Rehabilitation. She holds a patent for urine metabolites

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